

PCOMPBIOL-D-25-01995

Title: Degeneracy in astrocytic potassium buffering: A minimal model capturing the interplay between local and long-range mechanisms

Authors: Aitakin Ezzati, Nariman Kiani, Antonov Ivanov, Viktor Jirsa, Damien Depannemaecker, Christophe Bernard

Overview

Ezzati and colleagues present a study on the fundamental mechanism of astrocytic potassium buffering, revealing a potential degeneracy that could account for astrocytes' robust ionic homeostasis. The combination of experimental patch-clamp data with mathematical modeling is well aligned with the scope of *PLOS Computational Biology*. However, in its current form, the manuscript falls substantially short of the journal's standards in clarity, structure, biophysical interpretability, and the rigor of its claims.

While the authors repeatedly emphasize "minimality" and "degeneracy," these concepts are not developed with sufficient conceptual precision or convincingly grounded in biophysical reality. The manuscript suffers from severe issues in organization, figure presentation, mathematical formulation, and interpretation of results. As a consequence, the main claims—particularly those concerning degeneracy and homeostatic robustness—are not adequately supported, nor clarified for their biophysical (molecular) interpretation.

I recommend rejection at this time, while advising these authors to resubmit a substantially revised version in the future or to redirect their work to a more technical journal, such as, for example, the *Journal of Computational Neuroscience*, where the emphasis on formalism and exploratory modeling may be more appropriate for the current manuscript's version.

Detailed suggestions for editing

How can we address this?
Bifurcation analysis?

we can make a framework/pipeline and say from the gliocentric point of view?

The study is presented as a work aimed at establishing a "theoretical framework," yet, I would argue that the current implementation is closer to a phenomenological or exploratory model with strong assumptions embedded ad hoc (e.g., sigmoidal gating of spatial buffering, lumped K⁺ siphoning, limited ion species...). Moreover, the notion of "two levels of degeneracy" (cellular and functional) is intriguing, but it is introduced without a clear operational definition or experimental validation strategy.

I use the thesis introduction, talking about big picture of K from the gliocentric point of view

Introduction. The introduction provides a broad and largely accurate overview of astrocytic potassium buffering and its relevance to neuronal excitability and pathology. However, it lacks focus and narrative progression, reading more like an extended literature survey than a tightly argued rationale for the specific modeling choices in the study. Key concepts—spatial buffering, Kir channels, gap junctions, and degeneracy—are introduced, but not clearly connected in a way that motivates the proposed model. Likewise, the concept of degeneracy is underdeveloped. What distinguishes degeneracy from parameter non-identifiability or model sloppiness? Why degeneracy should be expected (or meaningful) in astrocytic K⁺ buffering, given existing experimental constraints. How degeneracy can be experimentally validated or falsified in this context.

from literature we see many works that address astrocytic buffering and the kinetics of K signal but there is no general review that gathers all these sporadic observations

Experimental Methods. How the experimental data constrain the model beyond membrane potential time courses is not articulated. Moreover, experimental variability, noise, or uncertainty, despite these being central to any claims about degeneracy, should be clarified.

Model Formulation

The exposition of the model lacks clarity and readability:

- Equations are poorly typeset, inconsistently numbered, and often difficult to parse.
- Notation is confusing and overloaded (e.g., multiple uses of similar symbols for concentrations, fluxes, and increments).
- Several variables are introduced without clear physical meaning or units, despite later claims of biophysical interpretability.

this I have to show the different version of the model, i tried (double sigmoid) also why is it like this?
- analyse the formulation in the open system;
spatial models, point model
- epistemology of the model

go through the scenarios that might lead to artifacts, and the argument regarding, with very small K_s we can consider I_{Kir} as I going out of the astrocyte, from another part not exposed to this ECS, but its, and it is due to the ionic and electrical coupling

Several modelling assumptions need to be supported, or explained:

- The introduction of a sigmoidal function to gate spatial buffering is not derived from first principles and is only loosely justified by experimental literature.

- The choice to use intracellular K^+ accumulation as a proxy for extracellular K^+ is debatable.
- The separation of intracellular K^+ into "local" and "syncytial" components without a clear physical grounding in the relevant spatial and temporal scales is prone to raising artifacts.

- for this we show the equilibrium; I-V curve adjustments the experiment of and bifurcation for time scale separation

Results

Model Dynamics and Experimental Validation

The main caveat of fitting a simplified model to experiments that include a (much) richer molecular biology is that model degeneracy may not mirror true biological degeneracy. This is particularly relevant in this study, where, in addition to the model simplifications, many parameters remain weakly constrained, leading to apparent degeneracy. After all, fitting membrane potential alone cannot account for the model's ionic dynamics. Indeed, several fitted parameters appear out of the reported physiological range (e.g., g_{Kir}). In this perspective, substantial effort should be devoted to interpreting the molecular biology of the different scenarios of degeneracy and linking them to the experimental setup and appropriate references.

Figures and Data Presentation

- Many figures lack units, properly labeled axes, or clear legends.
- Several figures could be combined into composite panels (e.g., Figs. 2-5, 6-7).
- There are often many similar traces across multiple figures, or many traces in individual panels, which hamper clarity.
- Several figures have panels arranged in a seemingly unstructured fashion, whereas they would be better shown on a grid.
- Very often figures report labels and titles that are not properly typeset.

Degeneracy Analysis

- The authors equate degeneracy with the existence of multiple parameter sets yielding similar outputs. This is a common property of nonlinear dynamical systems and optimization problems. The key missing aspect is that the authors do not distinguish between structural degeneracy and practical non-identifiability or parameter compensation.
- There is no experimental evidence that the different "degenerate" parameter regimes correspond to distinct astrocytic phenotypes or buffering pathways. Moreover, the robustness of degeneracy under perturbations (beyond the fitted conditions) is not systematically tested.
- The claim that degeneracy ensures extracellular K^+ homeostasis is problematic in light of the simplified nature of the ECS and bath coupling, the lack of competing homeostatic processes, and the overall absence of spatial structure in the model. Overall, it is unclear whether the observed extracellular K^+ degeneracy reflects a true biological principle or is simply a consequence of the model's imposed structure. These authors should thoroughly clarify and elaborate on this aspect.

so under spatiotemporal differences, we rely more on a specific mode; like DH and VH data

comparison with Gaute model claiming the point model version that includes the spatial extension effect; find point model justification in the lit b

higher K^+ where the degeneracy breaks, the bottleneck of a functional buffering unit, wallraff

meaning neurons nka and other exchanges

Discussion

this is for each phenotype compared to itself

- The authors argue that membrane kinetics can serve as a proxy for buffering efficiency, but this may not hold across different activity conditions and spatial scales.
- The model explicitly omits Na/K-ATPases, calcium-dependent pathways, pH regulation, volume changes, and spatial gradients within astrocytes or the ECS. I am not against the "minimal" modelling approach. Yet the absence of these mechanisms severely limits the interpretability of claims about homeostasis and robustness. Accordingly, the authors should provide a clear argument for why the observed "degeneracy" should be interpreted as a projection onto a lower-dimensional submanifold of a higher-dimensional (and possibly non-degenerate) physiological system.

I have to talk about how other mechanisms can be omitted and why it is good enough for this model and experiments, how can we be sure that these are the main mechanisms governing the buffering, make a deep research in intro and discussion based on the lit